



Research article

Irradiation of 125I seeds blocks glycolysis in pancreatic cancer by inhibiting KLF5 m6A methylation through the suppression of RBM15

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ABSTRACT

This paper investigated whether 125I seed irradiation for pancreatic cancer treatment was mediated through the RBM15/KLF5 pathway in glycolysis. The study collected peripheral blood from pancreatic cancer patients, and detected the expression of RBM15 and KLF5 expression in the serum of pancreatic cancer patients before and after 125I seed irradiation. An *in vitro* study was conducted to investigate the effects 125I seed irradiation on the malignant behavior and glycolysis of pancreatic cancer cells. The underlying mechanisms were thoroughly examined through a series of logical experiments, including Western blot analysis, Dot-blot experiment, methylated RNA immunoprecipitation assay, and RNA pull down assay. A xenograft tumor model in nude mice was established and treated with 125I seed irradiation, which was employed to research the *in vivo* effect and mechanism of 125I seed irradiation for pancreatic cancer. The overexpressed RBM15 and KLF5 in serum of pancreatic cancer patients were reduced after 125I seed treatment. 125I seed treatment impaired pancreatic cancer cell proliferation and invasion; enhanced apoptosis; attenuated glycolysis; and reduced RBM15 and KLF5 expression. RBM15 overexpression partially reversed these influences of 125I seed treatment on pancreatic cancer cells. RBM15 was capable of increasing KLF5 expression, which might be achieved by promoting m6A methylation of KLF5. *In vivo*, 125I seed treatment blocked the growth of pancreatic cancer cells and decreased RBM15 and KLF5 expression in xenograft tumor, whereas RBM15 overexpression abolished these effects. 125I seed irradiation suppressed glycolysis in pancreatic cancer by inhibiting KLF5 m6A methylation through down-regulation of RBM15. This discovery established a solid foundation for the use of in the treatment of pancreatic cancer.

1. Introduction

Pancreatic cancer is a highly aggressive malignant tumor, which is also the malignant tumor with the worst prognosis. The 5-year survival rate of patients with advanced pancreatic cancer is only 5 %, and even worse, the median survival of untreated patients with pancreatic cancer is only 4 months [1]. Due to the insidious early symptoms, the majority of patients with pancreatic cancer are already in the advanced stage when they are found for the first time, and thus lose the opportunity of curative surgery [2,3].

125I seed irradiation has been widely used in the treatment of

clinical entities, such as lung, thyroid and pancreatic cancers [4–6]. As a low-dose and brachytherapy method, 125I seed irradiation has achieved favorable local control in the treatment of entities, especially those that cannot be surgically excised [7]. 125I seed irradiation can ionize water molecules by continuously releasing low doses of X-rays and γ -rays, which in turn produce free radicals that can lead to DNA double-strand breaks and ultimately promote tumor cell apoptosis [1]. 125I seed irradiation is a minimally invasive and effective invasive treatment. Compared with conventional external radiotherapy, 125I seed irradiation has the advantages of fewer complications, less trauma, higher dose in the focal area, and less exposure of normal tissues [8]. 125I seed

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irradiation has achieved remarkable results in the clinical treatment of pancreatic cancer, but its mechanism of action in treating pancreatic cancer has not been fully clarified. It is well known that the high aggressiveness and rapid progression of pancreatic cancer are associated with a high rate of glycolysis [9]. Glycolysis, also known as the “Warburg effect”, can provide sufficient energy and nutrients to support the malignant behaviors of tumor cells, such as rapid proliferation and invasion [10]. Whether 125I seed irradiation exerts its inhibitory effect on pancreatic cancer by regulating glycolysis has not been clearly elucidated. Thus, this work researched the influence of 125I seed irradiation on glycolysis in pancreatic cancer.

Kruppel-like factor 5 (KLF5) plays pivotal role in modulating cell differentiation, proliferation and survival [11]. It is a newly discovered transcription factor that regulates glycolysis and lactate metabolism [12–14]. Data have also shown that KLF5 is associated with the prognosis of radiotherapy in cancer patients [15]. Although high KLF5 expression has recently been implied to be associated with pancreatic cancer [16], the available data are still very limited. The m6-methyladenosine (m6A) methylation modification refers to the replacement of one hydrogen in the amino group attached to the carbon atom 6 of adenine by a methyl group, which mainly functions to stabilize mRNA and promote translation [17]. The modification of m6A is widely involved in the regulation of tumor progression by modulating the biological behaviors of tumor cells, such as drug resistance, immune escape, and glycolysis [18]. In pancreatic cancer, dynamic changes in m6A methylation modification can drive tumor progression by promoting tumor cell adaptation to the hypoxic microenvironment, enhancing tumor cell glycolysis and chemoresistance, which in turn leads to poor prognosis [19,20]. RNA-binding motif protein15 (RBM15) functions to regulate m6A methylation of multiple genes, which in turn controls cell proliferation, apoptosis and disease progression [21,22]. It is also an oncogene in pancreatic cancer, and drives pancreatic cancer progression via m6A methylation modification [23,24]. Thus, this study initially researched whether 125I seed irradiation treated pancreatic cancer by regulating the RBM15-mediated KLF5 m6A methylation. This article will provide important clues and rationale for the clinical management of pancreatic cancer with 125I seed irradiation treatment.

2. Materials and methods

2.1. Clinical peripheral blood collection

Peripheral blood samples were collected from 40 patients with advanced pancreatic cancer and 40 healthy volunteers. Prior to the collection of blood samples, all participants were informed in detail, and voluntarily enrolled in the study. Then they voluntarily signed a written informed consent form. This study was ratified by the Ethics Committee of General Hospital of Ningxia Medical University, which complied with the Helsinki Declaration.

The inclusion criteria for the 40 patients with advanced pancreatic cancer were as follows: (1) confirmation of pancreatic cancer through pathologic diagnosis; (2) Stage III or IV patients who were not eligible for surgical resection; (3) absence of systemic infection; (4) absence of severe coagulation disorders in the peripheral blood; and (5) receipt of 125I treatment. The exclusion criteria included: (1) vital organ dysfunction or presence of other cancers; (2) prior receipt of other anti-cancer treatments; (3) incomplete 125I treatment or severe intolerance; (4) history of prior radiotherapy; and (5) presence of severe ascites or blood disorders. The 40 healthy volunteers had no blood disorders, no infectious diseases, and no vital organ dysfunction. The age range of the 40 patients with advanced pancreatic cancer was 55.3 ± 5.9 years, including 26 male patients and 14 female patients. The age range of the 40 healthy volunteers was 56.1 ± 5.1 years, including 26 male healthy volunteers and 14 female healthy volunteers. There was no statistically significant difference in age range and sex ratio between the 40 patients with advanced pancreatic cancer and the 40 healthy volunteers.

This study enrolled 40 patients with pancreatic cancer and 40 healthy volunteers. Power analysis was performed on the sample size. Using G*Power software, post-hoc power analysis was conducted with assuming $\alpha = 0.05$, effect size $d = 0.6$ (medium effect), and a sample size of 40. The formula for power analysis was as follows: $1 - \beta = \Phi\left(\frac{\sqrt{n} \cdot \%20d}{\sqrt{2}} - z_{1-\alpha/2}\right)$. As a result, the test efficacy reached 82 %. Thus, the sample size in this study was sufficient to detect statistical significance.

2.2. Overall survival analysis

The expression of RBM15 and KLF5 in pancreatic cancer patients was analyzed, along with its association with the overall survival of pancreatic cancer patients using the Kaplan-Meier Plotter online tool.

2.3. Cell lines and culture

Normal human pancreatic duct epithelial cell line (HPDE) and human pancreatic cancer cell lines (BxPC-3, SNU-213, and HPAC) were all provided by Hongshun Biotechnology (Shanghai, China). These cell lines were individually cultured in Dulbecco's Modified Eagle Medium (DMEM; Beyotime, Shanghai, China) with 10 % fetal bovine serum (FBS; Beyotime, Shanghai, China) under a 37 °C, 5 % CO₂ environment.

2.4. Cell transfection

Short interfering RNA (siRNA) against RBM15, siRNA against KLF5, and the corresponding siRNA negative control (NC), as well as RBM15 overexpression vectors, KLF5 overexpression vectors, and the corresponding NC vectors were all obtained from GeneChem (Shanghai, China). BxPC-3 cells in serum-free DMEM were separately transfected with these siRNAs or vectors using Lipofectamine 3000 (Thermo Fisher Scientific, San Jose, CA USA). Post transfection, BxPC-3 cells were grown in DMEM with 10 % FBS under a 37 °C, 5 % CO₂ environment.

2.5. Cell treatment

BxPC-3 cells were treated with 125I seed irradiation for 72 h at doses of 0.4 mCi and 0.8 mCi [25,26], respectively. The 125I seed was commercially provided by Beijing Atom and High Technology Industry (Beijing, China). The *in vitro* irradiation setup was as follows: a total of eight 125I seeds were wrapped evenly around a circumference (30 mm in diameter), while one 125I seed was placed in the center. The irradiation distance was 3 mm. Further, 3-deazaadenosine (3-DAA) (Med-ChemExpress, Shanghai, China), known as a broad-spectrum global methylation inhibitor, was used to treat BxPC-3 cells at a dose of 50 μM.

2.6. Cell counting kit-8 (CCK-8) assay

BxPC-3 cells were inoculated into 96-well plates, with 5000 cells in 100 μL of DMEM containing 10 % FBS in each well. After being treated for 24, 48 and 72 h under the relevant conditions, the cells were exposed to 10 μL of CCK-8 solution (Zeye Biotechnology, Shanghai, China) for 2 h at 37 °C. The plates were shaken for 30 min, and subsequently, the optical density (OD) value of each well was measured at 450 nm under a microplate reader (Biotek, Winooski, VT, USA).

2.7. Transwell invasion experiment

BxPC-3 cells were dispersed in serum-free DMEM at a concentration of 1×10^5 cells/mL. A 200 μL volume of the cell suspension was plated into the upper chamber, which had been pre-coated with Matrigel. In the lower chamber, 500 μL of DMEM with 20 % FBS was added. Cells were incubated for 24 h under a 37 °C, 5 % CO₂ environment. The invasion

cells were sequentially immobilized by 4 % paraformaldehyde (Yubo Biotechnology, Shanghai, China) and stained by 0.1 % crystal violet (Yubo Biotechnology, Shanghai, China) for 10 min, respectively. The invasion cells were placed under an inverted microscope (Olympus, Tokyo, Japan) for observation and counting.

2.8. Terminal transferase uridyl nick end labelling (TUNEL) assay

BxPC-3 cells in different groups were treated with 4 % paraformaldehyde for 15 min for fixation. After being exposed to 0.1 % Triton X-100 (Yubo Biotechnology, Shanghai, China) for 15 min, 50 μ L of TUNEL working solution (Zeye Biotechnology, Shanghai, China) was employed to treat cells for 1 h at 37 °C in darkness. Nucleus staining was accomplished by 5 min treatment with 4', 6-diamidino-2-phenylindole (DAPI; Yubo Biotechnology, Shanghai, China). The apoptotic cells were counted under a fluorescence microscopy (Olympus, Tokyo, Japan).

2.9. Extracellular acidification rate (ECAR), lactate production and intracellular glucose uptake

For ECAR assay, BxPC-3 cells (1×10^5 cells) were seeded into 6-well plates, and treated with 1 mL DMEM containing 10 % FBS, 10 mmol/L glucose (Solarbio, Beijing, China), 1 μ mol/L oligomycin (Solarbio, Beijing, China) and 100 mmol/L 2-deoxy-D-glucose (Solarbio, Beijing, China) for 24 h under a 37 °C, 5 % CO₂ environment. ECAR was assayed by CellTiter-Glo analysis. Regarding lactate production detection, BxPC-3 cells were grown in 24-well plates with 2×10^5 cells/mL. After 24 h of treatment, lactate content in the culture medium was tested using a Lactate Content Test Kit (Biolab Technology, Beijing, China). In terms of intracellular glucose uptake, BxPC-3 cells were seeded into 6-well plates (1×10^5 cells in 1 mL medium) for intracellular glucose uptake analysis. The cells were exposed to high-glucose DMEM for 24 h in a 37 °C, 5 % CO₂ environment. Subsequently, the cells were collected, and the intracellular glucose levels were measured using a Glucose Assay Kit (AmyJet Scientific, Wuhan, China).

2.10. Dot-blot experiment

Total RNA in BxPC-3 cells was extracted using Trizol reagent (Biolab Technology, Beijing, China). The total RNA weighing 100 ng was collected and then transferred onto a polyvinylidene fluoride (PVDF; Beyotime, Shanghai, China) membrane. After being crosslinked by ultraviolet (UV) irradiation, the membrane was treated with 5 % skimmed milk (Beyotime, Shanghai, China) for 1 h. The m6A methylation antibody (1:1000, ab208577, Abcam, Cambridge, UK) was applied to the membrane and allowed to incubate for 12 h at 4 °C, followed by treatment with a horseradish peroxidase (HRP)-labeled secondary antibody (1:2000, ab205718, Abcam, Cambridge, UK) for 1 h at room temperature. Finally, the dots were developed using Chemilum HRP Substrate (Beinuo Biotechnology, Shanghai, China). Furthermore, an equal amount of total RNA was dropped onto another PVDF membrane. Following crosslinking, the membrane was stained for 1 h with a 0.02 % methylene blue solution (Solarbio, Beijing, China). Ribonuclease-free water was used to rinse the membranes for 2 h. The dots were captured on by a camera.

2.11. Methylated RNA immunoprecipitation (Me-RIP) assay

The impact of RBM15 on KLF5 m6A methylation was investigated using a Magna MeRIP m6A Kit (Millipore Sigma, MA, USA). BxPC-3 cells were collected for total RNA extraction using Trizol reagent. Total RNA was immunoprecipitated with magnetic beads coated with anti-m6A antibody. The enriched KLF5 m6A RNA fragments were eluted and then qualified by real-time quantitative reverse transcription-polymerase chain reaction (qRT-PCR).

2.12. Actinomycin D treatment

BxPC-3 cells from different groups were treated with 2 μ g/mL actinomycin D (Solarbio, Beijing, China) for 0 min, 60 min, 120 min, 180 min and 240 min, respectively. At each time point, BxPC-3 cells were gathered for total RNA extraction using Trizol reagent. The level of KLF5 mRNA at each time point was assayed by qRT-PCR.

2.13. RNA pull down assay

This study generated biotin-labeled RNA by using the RiboTM RNAmix-T7 biotin-labeled transcription kit (Ribobio, Guangzhou, China). Subsequently, the biotin-labeled RNA was incubated with the streptavidin-labeled magnetic beads. BxPC-3 cells were lysed in radio-immuno precipitation assay (RIPA) lysis buffer (containing protease inhibitor) (Beyotime, Shanghai, China) for total protein extraction. The cell lysate was gathered and then incubated with the RNA-binding magnetic beads. After eluting the RNA-bound molecules, the RBM15 protein in the eluent was detected by Western blot.

2.14. Xenograft tumor experiment

BALB/c nude mice (male, 5-week old, $n = 18$) were supplied by KaiXue Biotechnology (Shanghai, China). All mice were maintained in a specific pathogen-free room at 22 °C with a 12-h light/night cycle. Mice were allowed free access to water and food. The animal experiment had been ratified by the Animal Ethics Committee of General Hospital of Ningxia Medical University.

The 18 mice were randomly divided into the Model group ($n = 6$), the 125I group ($n = 6$), and the 125I + RBM15 group ($n = 6$). Mice of the Model group and the 125I group were subcutaneously injected with BxPC-3 cells. The mice in the 125I + RBM15 group were injected with BxPC-3 cells transfected with lentivirus-based RBM15 overexpression vectors. Each mouse was injected with 1×10^6 cells in 100 μ L of phosphate buffer saline. When the diameter of the xenograft tumor reached 1 cm, 125I seeds (with a dose of 0.8 mCi) were implanted into the center of the xenograft tumor for mice in both the 125I group and the 125I + RBM15 group. Tumor volume was measured every seven days using the formula $(\text{length} \times \text{width}^2)/2$. After 28 days, mice were euthanized, and subcutaneous xenograft tumors were obtained and weighted. The xenograft tumors were preserved at -80 °C.

2.15. qRT-PCR

Total RNA in peripheral blood serum, cells, and xenograft tumors was isolated using Trizol reagent. The reverse transcription kit was purchased from Biolab Technology (Beijing, China), which was adopted to perform the reverse transcription reaction with total RNA in order to synthesize cDNA templates. For PCR analysis, an ABI 7500 Real Time PCR system (Applied Biosystems, Carlsbad, CA, USA) was utilized, with the following parameters were set: 95 °C for 1 min; 40 cycles of 95 °C for 15 s, 60 °C for 15 s, 72 °C for 45 s; followed by 70 °C for 5 min. β -actin and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) were used as controls for RBM15 and KLF5, respectively. The relative expression of RBM15 and KLF5 was determined using the $2^{-\Delta\Delta Ct}$ method.

2.16. Western blot

Total proteins were isolated from peripheral blood serum, cells, and xenograft tumors using RIPA lysis buffer (containing protease inhibitor). The content of total proteins was then assayed using a BCA assay kit (Biolab Technology, Beijing, China). By performing electrophoresis, the total proteins were transferred onto PVDF membranes and subsequently blocked with 5 % skimmed milk for 1 h. Following this, the proteins were probed with primary antibodies (including anti-RBM15 [1:1000, ab300467, Abcam, Cambridge, UK], anti-KLF5 [1:1000, ab137676,

Abcam, Cambridge, UK], anti- β -actin [1:1000, ab8227, Abcam, Cambridge, UK], and anti-GAPDH [1:1000, ab9485, Abcam, Cambridge, UK] for 12 h at 4 °C, and then with HRP-labeled goat anti-rabbit secondary antibody (1:2000, ab7085, Abcam, Cambridge, UK) for 2 h at room temperature. After being treated with enhanced chemiluminescence reagent (Biolab Technology, Beijing, China), the visualized blots were analyzed using with the Image J software (NIH, Bethesda, MD, USA). β -actin and GAPDH were respectively used as the controls for RBM15 and KLF5, respectively.

2.17. Immunohistochemistry

Xenograft tumors were fixed in 4 % paraformaldehyde for 24 h, and then embedded in paraffin. After being sliced, the tissue sections with a thickness of 4 μ m were deparaffinized with xylene, and then rehydrated with gradient alcohol. H_2O_2 solution (3 %) was adopted to treat the sections for 15 min, followed by 0.01 M citric acid buffer (Beyotime, Shanghai, China) boiling for 10 min. Normal goat serum (Biolab Technology, Beijing, China) was added to block the sections for 30 min. The sections were separately probed with anti-RBM15 (1:100, ab300467, Abcam, Cambridge, UK) and anti-KLF5 (1:100, ab137676, Abcam, Cambridge, UK) for 12 h at 4 °C, followed by incubation with HRP-labeled goat anti-rabbit secondary antibody (1:200, ab7085, Abcam, Cambridge, UK) for 30 min. In a dark environment, the sections were stained with 3,3'-diaminobenzidine (Biolab Technology, Beijing, China) for 5 min. After hematoxylin (Biolab Technology, Beijing, China) restaining, the sections were washed, dried, and observed under an inverted microscope (Olympus, Tokyo, Japan). The expression of

RBM15 and KLF5 proteins in the tissue sections was shown as brown particles.

2.18. Statistical analysis

All data (in the form of mean $\pm$ standard deviation) were statistically analyzed by GraphPad Prism 8 (GraphPad Software, San Diego, CA, USA). Before statistical analysis, data were tested to be normally distributed by Shapiro-Wilk test. Pairwise comparisons were conducted using a two-tailed paired Student's *t*-test. One-way analysis of variance, followed by post-hoc Tukey's test, was used for multiple comparisons. *P* < 0.05 was the threshold for a statistically significant difference.

3. Results

3.1. RBM15 and KLF5 were overexpressed in pancreatic cancer, but 125I treatment reduced it

This study began by monitoring the expression of RBM15 and KLF5 in the peripheral blood serum of pancreatic cancer patients and healthy volunteers. Higher mRNA and protein expression levels were observed for RBM15 and KLF5 in the peripheral blood serum of pancreatic cancer patients compared to those of healthy volunteers (*P* < 0.001) (Fig. 1A and B). In these pancreatic cancer patients, the expression of RBM15 and KLF5 in peripheral blood serum was dramatically reduced after 125I treatment (*P* < 0.001) (Fig. 1C and D). Based on the analysis of online database, high RBM15 and KLF5 expression was associated with poor overall survival of pancreatic cancer (Fig. 1E). Based on this data, it has

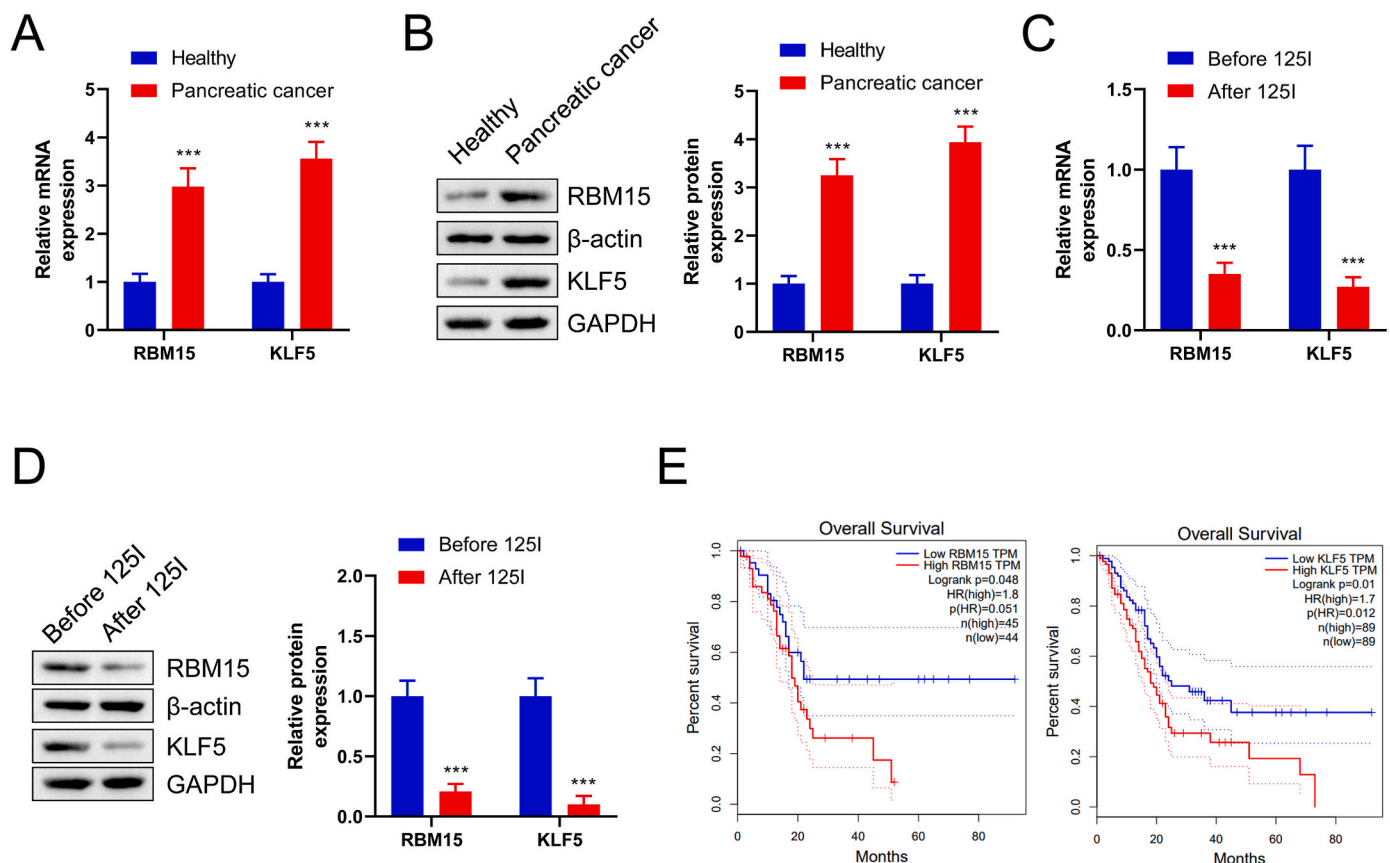


Fig. 1. 125I treatment down-regulated the overexpressed RBM15 and KLF5 in pancreatic cancer patients (A, B) qRT-PCR and Western blot revealed up-regulation of RBM15 and KLF5 in the peripheral blood serum of pancreatic cancer patients, with statistically significant difference. *n* = 40. ****P* < 0.001 vs. healthy volunteers. (C, D) qRT-PCR and Western blot showed the suppression of 125I treatment on RBM15 and KLF5 in the peripheral blood serum of pancreatic cancer patients, with statistically significant difference. *n* = 40. ****P* < 0.001 vs. before 125I treatment. (E) Analysis of Kaplan-Meier Plotter revealed that high RBM15 and KLF5 expression was associated with poor overall survival of pancreatic cancer patients.

been suggested that 125I could potentially be used as a treatment for pancreatic cancer by reducing the levels of RBM15 and KLF5. This hypothesis will be further investigated in the following studies.

3.2. 125I treatment impaired the proliferation of pancreatic cancer cells, reduced invasion and enhanced apoptosis

Normal human pancreatic duct epithelial cell line (HPDE) and human pancreatic cancer cell lines (BxPC-3, SNU-213 and HPAC) were subjected to qRT-PCR and Western blot to assay the expression of RBM15 and KLF5. Aberrantly higher mRNA and protein expression was found for RBM15 and KLF5 in BxPC-3, SNU-213 and HPAC cells, when matched to that in HPDE cells ($P < 0.001$) (Fig. 2A and B). Of the three pancreatic cancer cell lines, the expression of RBM15 and KLF5 in the BxPC-3 cell line showed an intermediate level, making it the subject of the following studies.

To investigate the effects of 125I treatment on the malignant behaviors of BxPC-3 cells, two different doses of 125I (0.4 mCi and 0.8

mCi, respectively) were administered to the cells. The proliferation, invasion and apoptosis of BxPC-3 cells were assessed using the CCK-8 assay (Fig. 2C), Transwell experiment (Fig. 2D), and TUNEL staining (Fig. 2E). It was discovered that both doses of 125I treatment significantly reduced BxPC-3 cell proliferation, invasion, and increased apoptosis ($P < 0.001$). Additionally, treatment with 125I at the 0.8 mCi dose was found to be more effective in inhibiting these harmful behaviors of BxPC-3 cells compared to the 0.4 mCi dose ($P < 0.001$).

3.3. 125I treatment reduced the expression of RBM15 and KLF5, and glycolysis in pancreatic cancer cells

After being treated with 125I (0.4 mCi and 0.8 mCi), BxPC-3 cells showed significantly lower levels of mRNA and protein for RBM15 and KLF5 ($P < 0.001$). Compared to BxPC-3 cells treated with 0.4 mCi 125I, those exposed to 0.8 mCi 125I exhibited even lower levels of mRNA and protein for RBM15 and KLF5 ($P < 0.001$) (Fig. 3A and B). Both doses of 125I treatment inhibited glycolysis in BxPC-3 cells, as it decreased

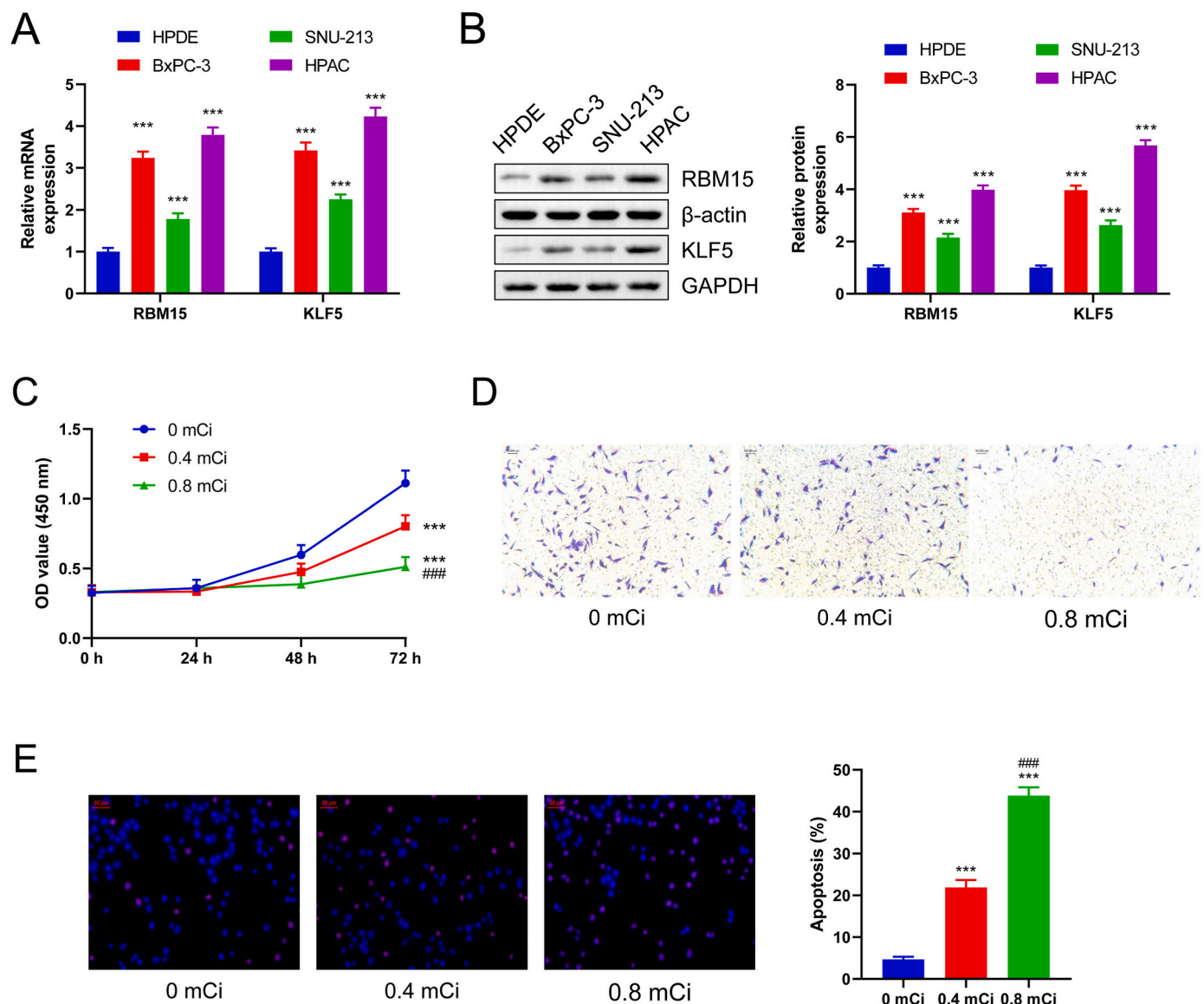


Fig. 2. 125I treatment suppressed pancreatic cancer cell proliferation, invasion and enhanced its apoptosis (A, B) qRT-PCR and Western blot revealed the up-regulation of RBM15 and KLF5 in human pancreatic cancer cell lines (BxPC-3, SNU-213 and HPAC), as compared to that in normal human pancreatic duct epithelial cell line (HPDE). The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. HPDE cell line. (C, D, E) The CCK-8 assay, Transwell experiment, and TUNEL staining revealed the suppression of 125I treatment on BxPC-3 cell proliferation and invasion, as well as the promotion on apoptosis. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. 0 mCi treatment. ### $P < 0.001$ vs. 0.4 mCi treatment.

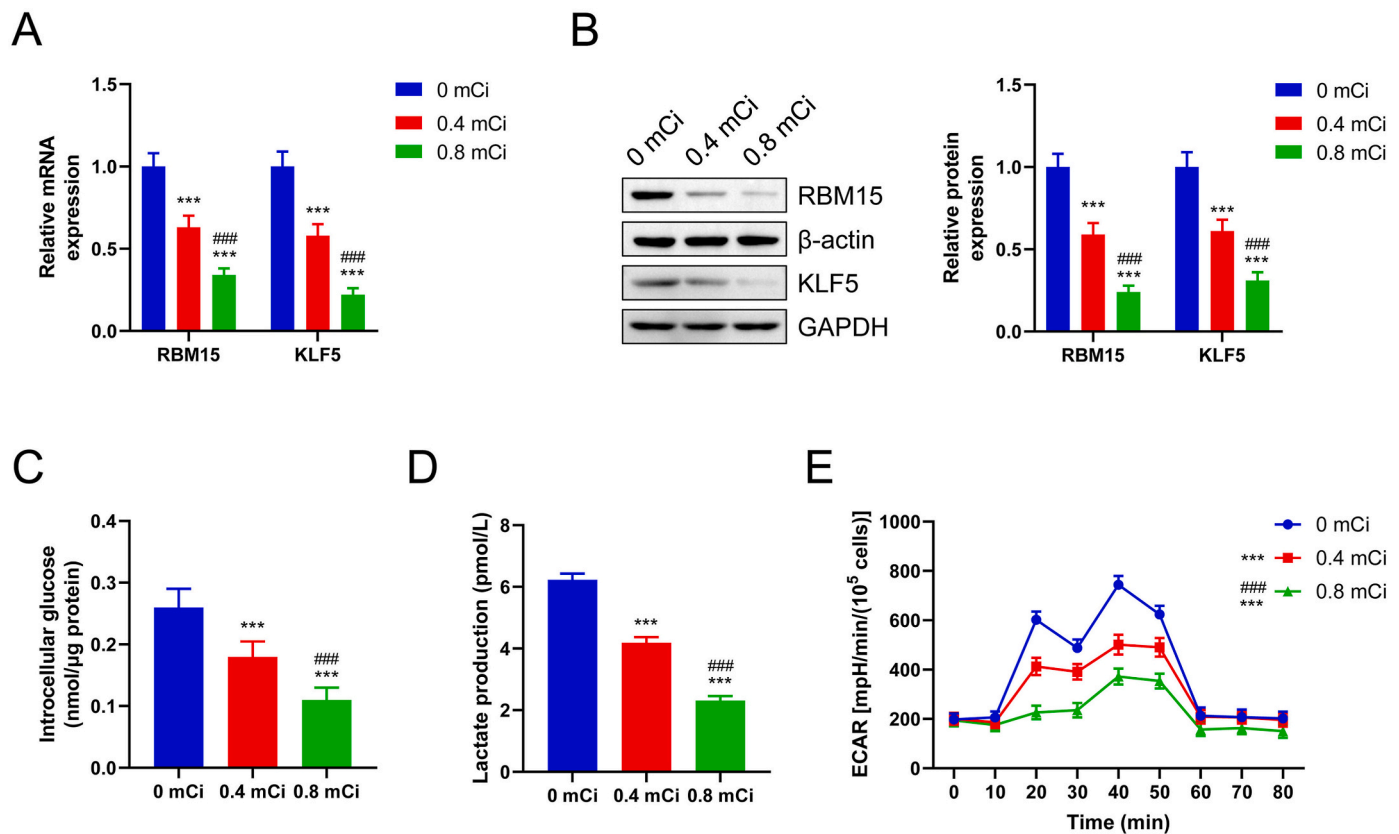


Fig. 3. 125I treatment inhibited the expression of RBM15 and KLF5, and glycolysis in pancreatic cancer cells (A, B) qRT-PCR and Western blot indicated that 125I treatment down-regulated RBM15 and KLF5 in BxPC-3 cells with statistically significant difference. $n = 3$. *** $P < 0.001$ vs. 0 mCi treatment. ### $P < 0.001$ vs. 0.4 mCi treatment. (C, D, E) 125I treatment blocked glycolysis in BxPC-3 cells, as demonstrated by the reduction of intracellular glucose uptake, lactate production and ECAR in BxPC-3 cells following 125I treatment. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. 0 mCi treatment. ### $P < 0.001$ vs. 0.4 mCi treatment.

intracellular glucose uptake, lactate production and ECAR in BxPC-3 cells ($P < 0.001$). The inhibition of glycolysis was more pronounced with the 0.8 mCi 125I treatment than with the 0.4 mCi 125I treatment ($P < 0.001$) (Fig. 3C, D, E). Therefore, 125I treatment was capable of suppressing the expression of RBM15 and KLF5, as well as glycolysis in pancreatic cancer cells.

3.4. RBM15 overexpression partially reversed the inhibition of 125I treatment on the malignant behavior of pancreatic cancer cells

siRNA targeting RBM15, siRNA NC, RBM15 overexpression vectors, and NC vectors were separately transfected into BxPC-3 cells. As depicted in Fig. 4A, transfection with RBM15 siRNA resulted in a significant reduction in RBM15 protein expression ($P < 0.001$), whereas transfection with the RBM15 overexpression vector led to a notable increase in RBM15 protein expression ($P < 0.001$). This indicated that RBM15 expression in BxPC-3 cells was effectively controlled through transfection.

With regard to functional experiments, CCK-8 assay (Fig. 4B), Transwell experiment (Fig. 4C), and TUNEL staining (Fig. 4D) were implemented to evaluate the viability, invasion and apoptosis of BxPC-3 cells. In comparison to control BxPC-3 cells, 125I treatment (the 125I group) or RBM15 siRNA transfection (si-RBM15 group) significantly attenuated BxPC-3 cell viability and invasion, but intensified its apoptosis ($P < 0.001$). On the contrary, transfection of the RBM15 overexpression vector transfection (the RBM15 group) markedly enhanced the viability and invasion of BxPC-3 cells, but reduced their apoptosis, as relative to control BxPC-3 cells ($P < 0.001$). In the 125I treatment group (the 125I group), a reduction in viability and invasion,

along with an increase in apoptosis, was found in BxPC-3 cells of the 125I + si-RBM15 group ($P < 0.001$). Conversely, BxPC-3 cell viability and invasion were increased and apoptosis was decreased in the 125I + RBM15 group, compared to the 125I group ($P < 0.001$). Thus, RBM15 overexpression was capable of counteracting the suppression of 125I treatment on the malignant behavior of pancreatic cancer cells. This suggested that 125I treatment inhibited the malignant behavior of pancreatic cancer cells by reducing RBM15 expression.

3.5. RBM15 overexpression abrogated the suppression of 125I treatment on KLF5 expression and glycolysis in pancreatic cancer cells

The expression of KLF5 and glycolysis in BxPC-3 cells from different treatment groups were evaluated by qRT-PCR (Fig. 5A), Western blot (Fig. 5B), and glycolysis assay (Fig. 5C, D, E). As matched to control BxPC-3 cells, those treated by 125I (the 125I group) or transfected by RBM15 siRNA (si-RBM15 group) possessed lower mRNA and protein expression for KLF5, and lower intracellular glucose uptake, lactate production and ECAR ($P < 0.001$). However, significantly higher levels of mRNA and protein expression for KLF5, along with increased intracellular glucose uptake, lactate production and ECAR, were observed in BxPC-3 cells transfected with the RBM15 overexpression vector (the RBM15 group), when referred to control BxPC-3 cells ($P < 0.001$). In contrast, BxPC-3 cells treated only with 125I (the 125I group) showed lower levels of lower mRNA and protein expression for KLF5 as well as decreased intracellular glucose uptake, lactate production and ECAR. This was observed in the 125I + si-RBM15 group ($P < 0.001$). Conversely, the 125I + RBM15 group exhibited opposite results with significantly higher levels of these parameters ($P < 0.001$). Therefore,

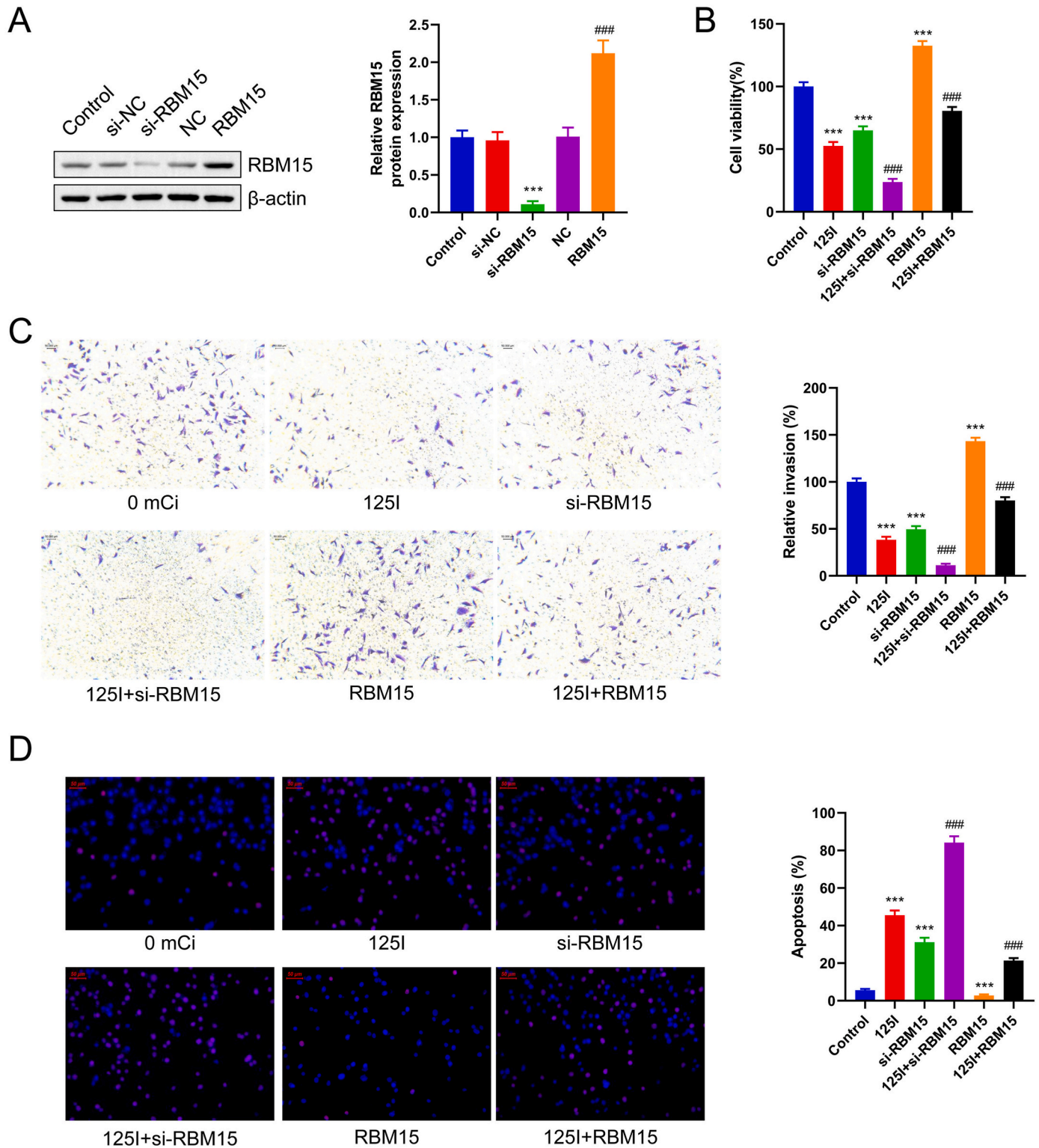


Fig. 4. RBM15 overexpression abrogated the suppression of 125I treatment on the malignant behavior of pancreatic cancer cells (A) Western blot illustrated that RBM15 expression in BxPC-3 cells was successfully modulated by transfection. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the si-NC group. ### $P < 0.001$ vs. the NC group. (B, C, D) According to CCK-8 assay, Transwell experiment, and TUNEL staining, the suppression of 125I treatment on BxPC-3 cell viability and invasion, as well as the promotion on its apoptosis, was abolished by RBM15 overexpression. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the Control group. ### $P < 0.001$ vs. the 125I group.

the enhancement of 125I treatment on KLF5 expression and glycolysis in pancreatic cancer cells was reversed by RBM15 overexpression, suggesting that 125I treatment could suppress KLF5 expression and glycolysis in pancreatic cancer cells by reducing RBM15 expression.

3.6. KLF5 overexpression counteracted the inhibition of 125I treatment on the malignant behavior and glycolysis of pancreatic cancer cells

KLF5 siRNA, KLF5 overexpression vectors, and their corresponding

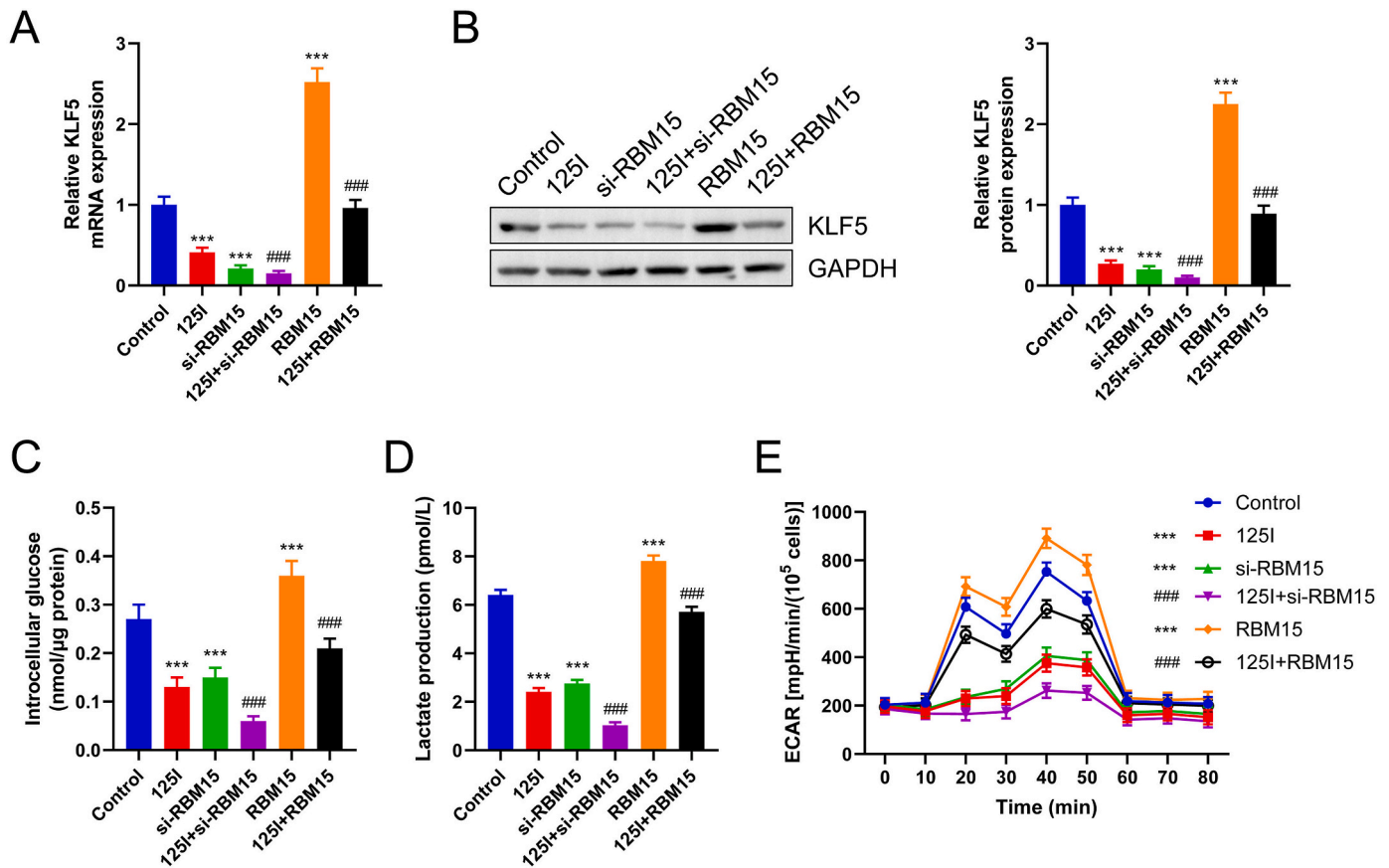


Fig. 5. RBM15 overexpression partially reversed the suppression of 125I treatment on KLF5 expression and glycolysis in pancreatic cancer cells (A, B) By qRT-PCR and Western blot, RBM15 overexpression in BxPC-3 cells partially reversed the suppression of 125I treatment on KLF5 expression in BxPC-3 cells, with statistically significant difference. $n = 3$. *** $P < 0.001$ vs. the Control group. ### $P < 0.001$ vs. the 125I group. (C, D, E) Glycolysis assay indicated that, RBM15 overexpression abolished the inhibition of 125I treatment on intracellular glucose uptake, lactate production and ECAR in BxPC-3 cells. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the Control group. ### $P < 0.001$ vs. the 125I group.

negative controls were individually transfected into BxPC-3 cells. The transfection of KLF5 siRNA significantly decreased KLF5 protein levels in BxPC-3 cells (the si-KLF5 group vs. the Control group) ($P < 0.001$), while transfection with the KLF5 overexpression vector notably increased KLF5 levels (the KLF5 group vs. the Control group) ($P < 0.001$) (Fig. 6A). Therefore, KLF5 expression was successfully modulated through transfection.

CCK-8 assay (Fig. 6B), Transwell experiment (Fig. 6C), TUNEL staining (Fig. 6D) and glycolysis assay (Fig. 6E, F, G) were carried out on BxPC-3 cells in each group. When referred to the Control group, lower cell viability and invasion, higher apoptosis, and lower intracellular glucose uptake, lactate production and ECAR were found in BxPC-3 cells of the 125I group and the si-KLF5 group ($P < 0.001$). However, contrasting data were found in BxPC-3 cells of KLF5 group compared to the Control group ($P < 0.001$). In contrast to the 125I group, BxPC-3 cells of the 125I + si-KLF5 group showed the reduced viability and invasion, the increased apoptosis, and decreased intracellular glucose uptake, lactate production and ECAR ($P < 0.001$). However, the cells in the 125I + KLF5 group exhibited increased viability and invasion, decreased apoptosis, and higher levels of intracellular glucose uptake, lactate production and ECAR compared to the cells in the 125I group ($P < 0.001$). These results indicated that overexpression of KLF5 reversed the inhibitory effects of 125I treatment on the aggressive behavior and glycolysis of pancreatic cancer cells. Thus, the 125I treatment was found to suppress the aggressive behavior and glycolysis of pancreatic cancer cells by reducing the expression of KLF5.

3.7. RBM15 enhanced m6A methylation of KLF5 in pancreatic cancer cells

The interaction between RBM15 and KLF5 was explored. RBM15 siRNA transfection of BxPC-3 cells caused a decrease in mRNA and protein expression for KLF5 (the si-RBM15 group vs. the si-NC group) ($P < 0.001$). In contrast, transfection of RBM15 overexpression vectors markedly increased intensified mRNA and protein expression of KLF5 compared to the NC group ($P < 0.001$) (Fig. 7A and B). A dot-blot experiment (Fig. 7C) and Me-RIP assay (Fig. 7D) were conducted to investigate the impact of RBM15 on KLF5 m6A methylation. As a result, silencing RBM15 in BxPC-3 cells significantly decreased KLF5 m6A methylation and KLF5 enrichment compared to the si-NC group ($P < 0.001$). Conversely, overexpression of RBM15 resulted in a significant increase in KLF5 m6A methylation and KLF5 enrichment compared to the NC group ($P < 0.001$).

In the actinomycin D treatment experiment, silencing RBM15 in BxPC-3 cells accelerated the degradation rate of KLF5 mRNA (the si-RBM15 group vs. the si-NC group) ($P < 0.001$), while overexpression of RBM15 attenuated it (the RBM15 group vs. the NC group) ($P < 0.001$) (Fig. 7E). The RNA pull down assay showed the pull-down effect of RBM15 on KLF5 (Fig. 7F). Finally, this work used 3-DAA (m6A methylation inhibitor) to treat BxPC-3 cells. Higher mRNA and protein expression of KLF5 was observed in BxPC-3 cells of the RBM15 group compared to the NC group ($P < 0.001$). After 3-DAA treatment, mRNA and protein expression of KLF5 were significantly decreased in the presence of RBM15 overexpression or not (the NC + 3-DAA group vs. the NC group; the RBM15 + 3-DAA group vs. the RBM15 group) ($P < 0.001$).

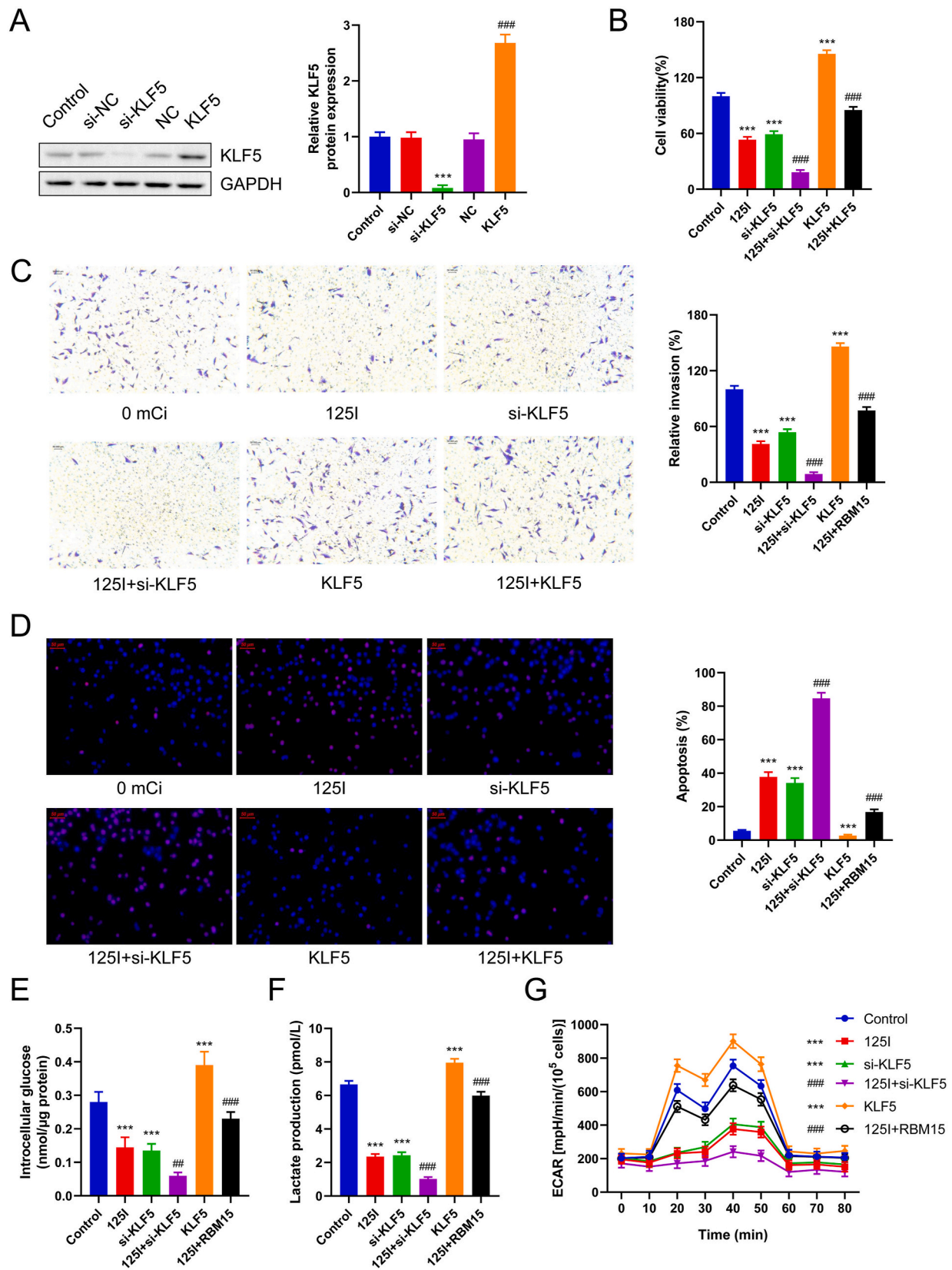


Fig. 6. KLF5 up-regulation abolished the repression of 125I treatment on the malignant behavior and glycolysis of pancreatic cancer cells (A) By Western blot, KLF5 expression in BxPC-3 cells was effectively modulated by transfection with statistically significant difference. $n = 3$. *** $P < 0.001$ vs. the si-NC group. ### $P < 0.001$ vs. the NC group. (B, C, D) The CCK-8 assay, Transwell experiment and TUNEL staining implied that, KLF5 up-regulation partially reversed the suppression of 125I treatment on BxPC-3 cell viability and invasion, and the promotion on its apoptosis. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the Control group. ### $P < 0.001$ vs. the 125I group. (E, F, G) By glycolysis assay, KLF5 up-regulation counteracted the repression of 125I treatment on glycolysis of BxPC-3 cells. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the Control group. ### $P < 0.001$ vs. the 125I group.

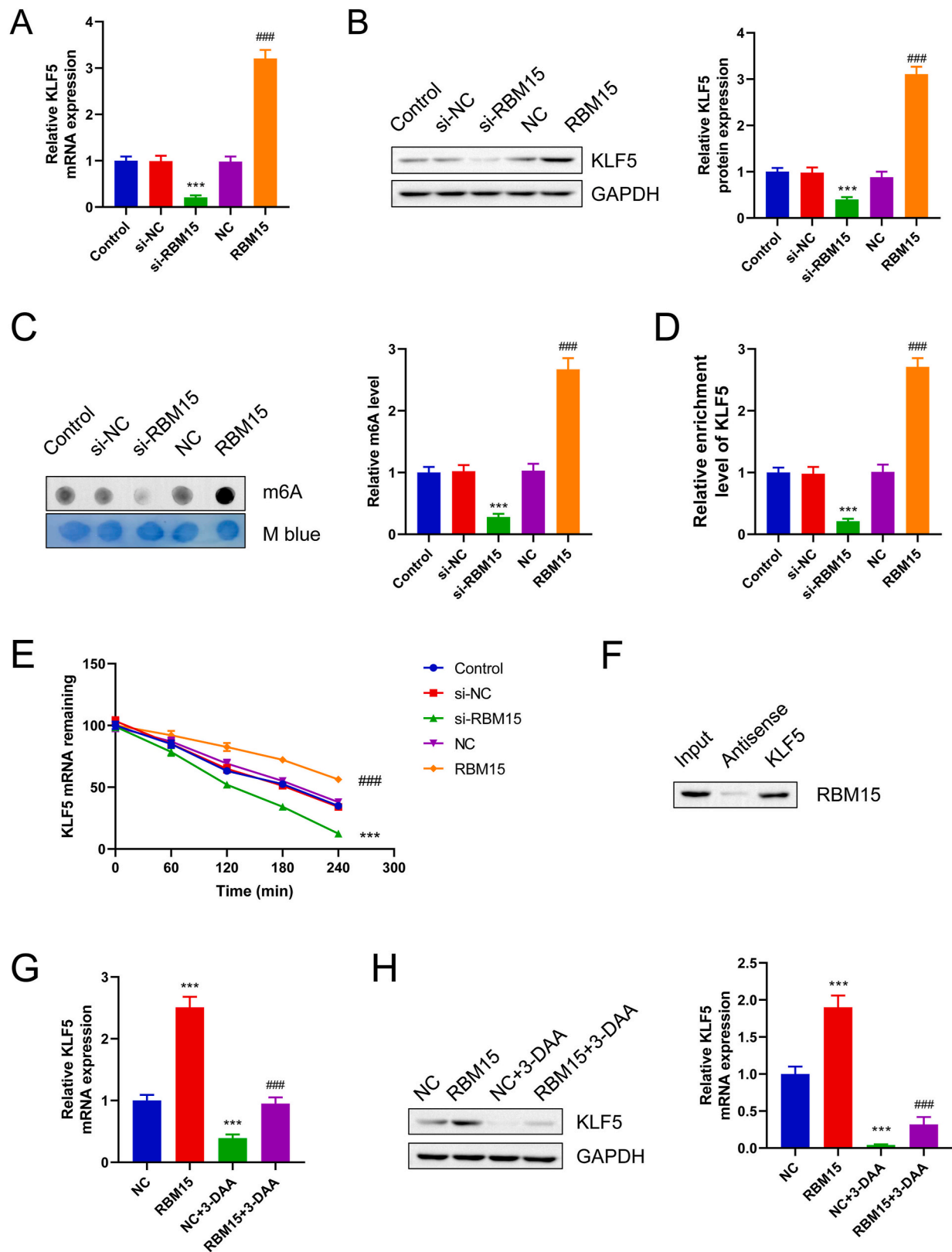


Fig. 7. RBM15 facilitated m6A methylation of KLF5 in pancreatic cancer cells (A, B) Through qRT-PCR and Western blot, RBM15 overexpression in BxPC-3 cells elevated mRNA and protein expression for KLF5, with statistically significant difference. $n = 3$. $^{**}P < 0.001$ vs. the si-NC group. $^{###}P < 0.001$ vs. the NC group. (C, D) Dot-blot experiment and Me-RIP assay illustrated that RBM15 overexpression intensified m6A methylation and enrichment of KLF5 in BxPC-3 cells. The difference was statistically significant. $n = 3$. $^{**}P < 0.001$ vs. the si-NC group. $^{###}P < 0.001$ vs. the NC group. (E) By actinomycin D treatment experiment, RBM15 overexpression attenuated the degradation rate of KLF5 mRNA in BxPC-3 cells, with statistically significant difference. $n = 3$. $^{**}P < 0.001$ vs. the si-NC group. $^{###}P < 0.001$ vs. the NC group. (F) RNA pull down assay revealed the pull-down effect of RBM15 on KLF5. (G, H) 3-DAA treatment blocked the promotion of RBM15 on mRNA and protein expression for KLF5. The difference was statistically significant. $n = 3$. $^{***}P < 0.001$ vs. the NC group. $^{###}P < 0.001$ vs. the RBM15 group.

(Fig. 7G and H). All of the data implied that RBM15 might promote KLF5 expression in pancreatic cancer cells by enhancing its m6A methylation.

3.8. RBM15 overexpression abolished the suppression of 125I treatment on the *in vivo* growth of pancreatic cancer cells

Through *in vivo* experiments, 125I treatment resulted in a significant decrease in xenograft tumor volume and weight compared to the Model group ($P < 0.001$). However, the suppressive effect of 125I treatment on the *in vivo* growth of BxPC-3 cells was negated by RBM15

overexpression. This was evidenced by the higher xenograft tumor volume and weight in the 125I + RBM15 group compared to the 125I group ($P < 0.001$) (Fig. 8A, B, C).

The expression of RBM15 and KLF5 in xenograft tumors from different groups was evaluated using qRT-PCR (Fig. 8D), Western blot (Fig. 8E), and immunohistochemistry (Fig. 8F). Lower expression of RBM15 and KLF5 was observed in xenograft tumors of the 125I group compared to the Model group ($P < 0.001$). Conversely, in contrast to the 125I group, xenograft tumors of the 125I + RBM15 group presented higher RBM15 and KLF5 expression ($P < 0.001$). Therefore, all the

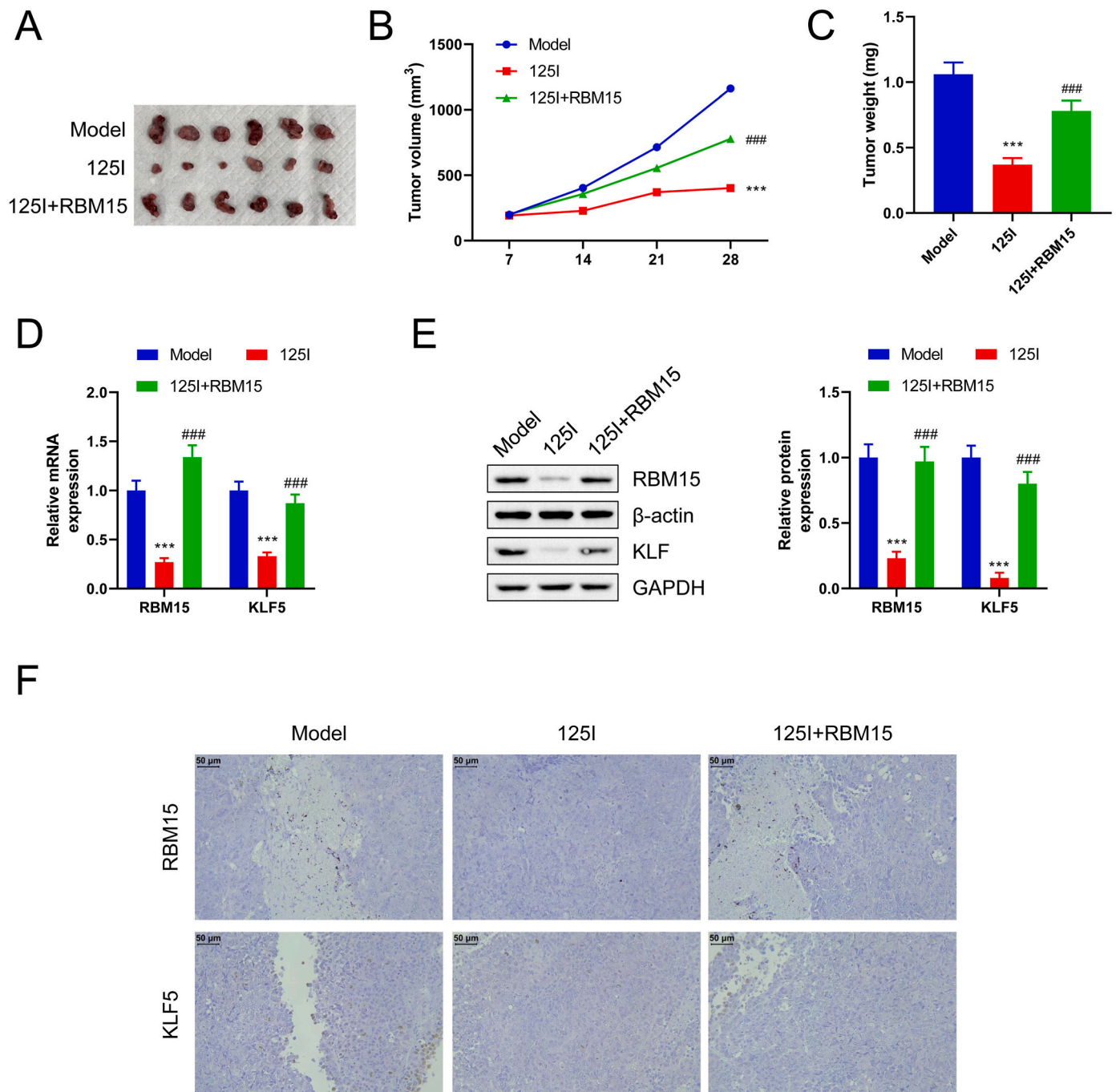


Fig. 8. RBM15 overexpression abrogated the repression of 125I treatment on the *in vivo* growth of pancreatic cancer cells (A, B, C) 125I treatment suppressed BxPC-3 cell growth *in vivo*, which was partially reversed by RBM15 overexpression. The difference was statistically significant. $n = 6$. *** $P < 0.001$ vs. the Model group. ### $P < 0.001$ vs. the 125I group. (D, E, F) Through qRT-PCR, Western blot and immunohistochemistry, 125I treatment down-regulated RBM15 and KLF5 expression in xenograft tumors, but RBM15 overexpression abolished the suppression of 125I treatment on the expression of KLF5 in xenograft tumors. The difference was statistically significant. $n = 3$. *** $P < 0.001$ vs. the Model group. ### $P < 0.001$ vs. the 125I group.

discoveries in this work implied that 125I treatment might suppress the progression of pancreatic cancer by reducing KLF5 m6A methylation via inhibiting RBM15 expression.

4. Discussion

This paper monitored the overexpressed RBM15 and KLF5 in patients with pancreatic cancer. High RBM15 and KLF5 expression in patients with pancreatic cancer was associated with poor overall survival. RBM15 has been identified as an oncogene in a wide range of human cancers [27–29]. For instance, in lung adenocarcinoma, the expression of RBM15 is abnormally increased, which not only promotes the malignant behavior of tumor cells, but also correlates with poorer overall survival of patients [30]. RBM15 is an oncogene in cervical cancer, and it drives tumor progression by enhancing invasion, proliferation and stem cell property of tumor cells [31]. In pancreatic cancer, high RBM15 expression has been reported to be associated with poor overall survival, disease-free survival, progression-free survival, and disease-specific survival [32]. Based on the available data, RBM15 silencing inhibits the *in vitro* proliferation and invasion of pancreatic cancer cells, and blocks the *in vivo* growth of pancreatic cancer cells in nude mice [23]. In fact, RBM15 silencing functions to promote macrophage infiltration, which alleviates disease progression of pancreatic cancer by enhancing macrophage phagocytosis of pancreatic cancer cells [24]. This study indicated that RBM15 acted as an oncogene in pancreatic cancer, and it enhanced malignant behavior of pancreatic cancer cells such as proliferation, invasion, and glycolysis. Moreover, RBM15 overexpression suppressed the apoptosis of pancreatic cancer cells. Thus, the findings in this paper were consistent with these above previous studies.

KLF5 has been found to exert oncogenic functions on a variety of human cancer by participating in multiple pathways [33–35]. It has been found to enhance metastasis in nasopharyngeal cancer and promote radioresistance in esophageal squamous cell carcinoma [36,37]. Conversely, KLF5 has oncogenic activity in cervical cancer, which overexpression inhibits invasion and migration of cervical cancer cells [38]. Thus, KLF5 plays different functions in different cancer types. In pancreatic cancer, ablation of KLF5 has been discovered to alter the immune tumor microenvironment, thereby enhancing tumor sensitivity to immunotherapy [39]. High KLF5 expression can accelerate the progression of pancreatic cancer by facilitating cell cycle progression, which is thus proposed to be an important prognostic biomarker for pancreatic cancer [40]. In this study, up-regulation of KLF5 in pancreatic cancer cells was demonstrated to intensify pancreatic cancer cell proliferation, invasion, and glycolysis, but mitigated its apoptosis. Conversely, silencing of KLF5 resulted in an opposite effect. Therefore, KLF5 was suggested to be an oncogene in pancreatic cancer in this work.

This study discovered, for the first time, that 125I seed irradiation could decrease the expression of RBM15 and KLF5 in patients with pancreatic cancer. Building on this groundbreaking discovery, *in vitro* experiments were then carried out by exposing pancreatic cancer cells to 125I seed irradiation. It was revealed that 125I seed irradiation repressed malignant behavior (such as proliferation, invasion, and apoptosis inhibition) and glycolysis of pancreatic cancer cells. At the same time, 125I seed irradiation down-regulated RBM15 and KLF5 in pancreatic cancer cells. Rescue experiments implied that overexpression of RBM15 and KLF5 partially reversed the suppression of 125I seed irradiation on malignant behavior and glycolysis of pancreatic cancer cells. Therefore, 125I seed irradiation was proposed to block disease progression of pancreatic cancer by down-regulating RBM15 and KLF5. 125I seed irradiation is a commonly employed clinical strategy for the treatment of solid tumors, especially for patients who cannot be surgically resected [41,42]. 125I seed irradiation significantly reduces the complications of conventional radiation therapy, and its short-range radiation advantage also greatly mitigates damage to normal tissues [43]. 125I seed irradiation has achieved satisfactory efficacy in the clinical management of pancreatic cancer [44,45]. The downside,

however, is that the molecular mechanisms by which 125I seed irradiation treat pancreatic cancer have still not been clearly elucidated. To the best of our knowledge, this was the first time that 125I seed irradiation was demonstrated to block pancreatic cancer progression by reducing RBM15 and KLF5.

RBM15 is well known as a key regulator of m6A methylation, which modulates disease progression by stabilizing multiple gene expression via promoting m6A methylation of these genes [32]. m6A methylation is the most abundant RNA modification with the function of stabilizing gene expression [46]. Several methylation regulators (including RBM15), along with m6A methylation of some genes, have been researched to be associated with survival of patients with pancreatic cancer [47]. As a m6A methylation regulator, RBM15 has also been associated with advanced pancreatic cancer [48]. However, the influence of RBM15 on m6A methylation of KLF5 has never been studied. KLF5, a glycolysis-promoting factor, is widely involved in tumor development and is proposed to be a potential target for cancer therapy [14,49]. Mutation in KLF5 has been noticed to be related to glycolysis and progression of pancreatic cancer [50]. Therefore, revealing the effect of RBM15 on KLF5 m6A methylation will help to deepen the understanding of the mechanism of pancreatic cancer progression. Initially, this study demonstrated the role of RBM15 in facilitating KLF5 m6A methylation. Therefore, it was possible that 125I seed irradiation inhibited glycolysis in pancreatic cancer by decreasing KLF5 m6A methylation through the suppression of RBM15. This discovery provided important insights into the molecular mechanisms of pancreatic cancer progression, which might provide effective clues and strategies for the treatment of pancreatic cancer. The results of the *in vivo* experiments further reinforced this conclusion. In terms of clinical significance, the promotion of RBM15 on KLF5 m6A methylation could serve as a molecular basis for the development of new strategies for the treatment of pancreatic cancer. Further, this paper provided a definitive molecular basis for 125I seed irradiation in the clinical treatment of pancreatic cancer.

There are limitations in the research. It would be highly clinically significant to monitor the dynamic changes of RBM15 and KLF5 in clinical samples over a period of time. However, this topic is currently difficult to perform in the clinical setting, which will be the focus of our future research. Moreover, in addition to RBM15 and KLF5, there are many other transcription factors and regulators in pancreatic cancer cells that may affect glycolysis and cell growth. Researching other transcription factors and regulators will help in the development of therapeutic strategies for pancreatic cancer. However, due to laboratory limitations, this issue cannot be performed at present, and it will be the focus of our future research.

The present paper initially revealed the suppression of 125I seed irradiation on the overexpressed RBM15 and KLF5 in pancreatic cancer. RBM15 and KLF5 were oncogenes in pancreatic cancer, which drove the malignant behavior and glycolysis of pancreatic cancer cells. 125I seed irradiation could suppress malignant behavior and glycolysis of pancreatic cancer cells, possibly by attenuating KLF5 m6A methylation via repressing RBM15. This study provided a clear molecular basis for the application of 125I seed irradiation in the clinical treatment of pancreatic cancer. Furthermore, the promotion of KLF5 m6A methylation by RBM15 provided a molecular basis for the development of new strategies for the clinical treatment of pancreatic cancer.

CRediT authorship contribution statement

Jianjun Song: Conceptualization. **Fuping Kang:** Formal analysis. **Xiaoning Chang:** Formal analysis. **Jing Wu:** Formal analysis. **Peng Zhang:** Formal analysis. **Xiaolong Wang:** Formal analysis. **Feng Wang:** Conceptualization.

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Declaration of competing interest

We all declare that we have no conflict of interest./There is no conflict of interest.

Data availability

Data will be made available on request.

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